Theory of Computation

CNF, GNF and Arden's Theorem

Lecture 40

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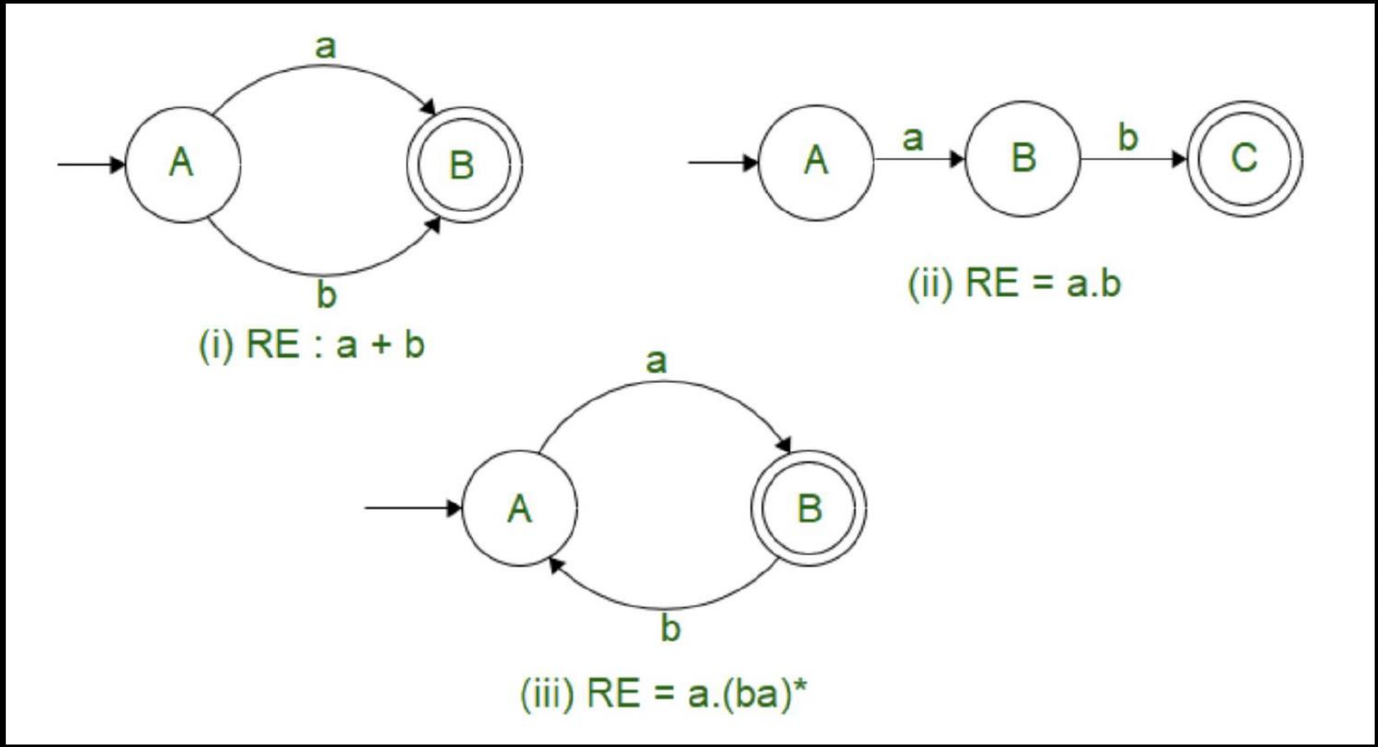
Arden's Theorem

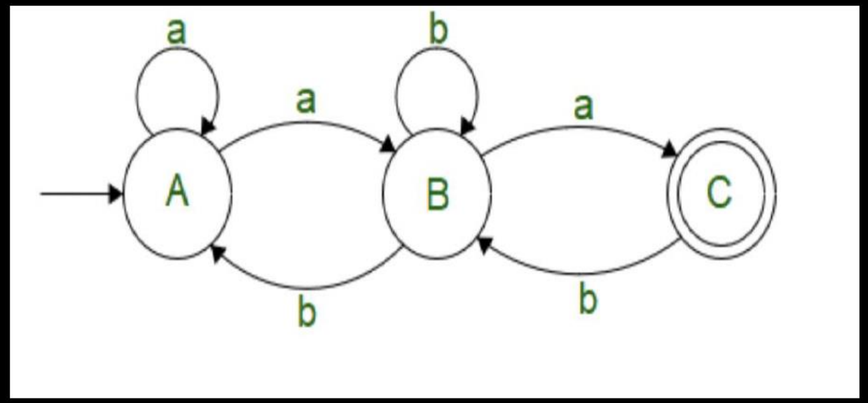
Arden's Theorem states that, if P & Q are two regular expressions over $\sum$, and if P does not contain "$\in$", then the following equation R given by

$R = Q + RP$ has a unique solution

$R = QP^*$

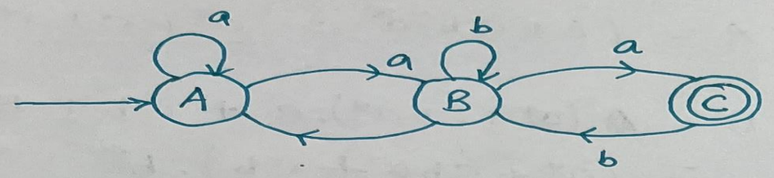
Directly we can resolve simple DFA/NFA into regular expression





For Complex DFA/NFA, we can use algorithmic approach to find regular expression(Arden’s Theorem)

(1)



$C = Ba \longrightarrow (1)$

$B = Aa + Bb + Cb \longrightarrow (2)$

$A = \varepsilon + Aa + Bb \longrightarrow (3)$

(1) $\rightarrow C = Ba = (Aa + Bb + Cb).a$

$= Aaa + Bba + Cba \longrightarrow (4)$

(2) $\rightarrow \underset{R}{B} = Aa + Bb + Cb = Aa + Bb + (Ba)b$

$= \underbrace{Aa}_{Q} + \underset{R}{B}\underbrace{(b + ab)}_{P}$

$\therefore B = Aa(b+ab)^* \longrightarrow (5)$

(3) $\rightarrow \underset{R}{A} = \varepsilon + Aa + Bb$

$= \varepsilon + Aa + (Aa(b+ab)^*).b$

$= \underset{Q}{\varepsilon} + \underset{R}{A}\underbrace{(a + a(b+ab)^*.b)}_{P} \longrightarrow (6)$

$A = (a + a(b+ab)^*.b)^* \longrightarrow (6)$

R.E (for state C, as it is final state)

$C = Ba = \underbrace{Aa(b+ab)^*}_{B}.a$

$= \boxed{(a + a(b+ab)^*b)^*.\ a(b+ab)^*.a}$

R.E

Ans

Chomsky Normal Form (CNF)

Definition: A CFG is in CNF if every rule has one of the forms:

$A \to BC$ (two non-terminals)

$A \to a$ (single terminal)

Also: $S \to \varepsilon$ if S is start symbol and $\varepsilon \notin L(G)$

Example Grammar → CNF

Original Rule	CNF Rule(s)	Reason
$S \to ABa$	$S \to AX_1$, $X_1 \to Ba$	3-body rule: split + terminal
$A \to aA \mid a$	$A \to T_1A \mid T_1$, $T_1 \to a$	Replace terminal in 2-body

Greibach Normal Form (GNF)

Definition: A CFG is in GNF if every rule has the form:

$A \rightarrow a\ \alpha$ (terminal, then ≥0 non-terminals)

Where a is a terminal and $\alpha \in (V)^*$ is a (possibly empty) string of non-terminals

Example Grammar → GNF

Original Rule	GNF Rule(s)	Note
S → AB	S → aBS'... (expand A)	A must start with terminal
A → aA | a	Already in GNF ✓	Starts with terminal a

MCQ: Chomsky Normal Form

Q1. Which rule is NOT in CNF?

A) $A \rightarrow BC$

B) $A \rightarrow a$

C) $A \rightarrow ABC$

D) $S \rightarrow \varepsilon$

Q2. To convert $A \rightarrow aBC$ to CNF, we introduce $T \rightarrow a$. The rule becomes:

A) $A \rightarrow TBC$

B) $A \rightarrow aBC$

C) $A \rightarrow TBC, T \rightarrow B$

D) $A \rightarrow TC$

MCQ: Greibach Normal Form

Q3. Which of the following IS in GNF?

A) $A \rightarrow BC$

B) $A \rightarrow aBC$

C) $A \rightarrow ABa$

D) $A \rightarrow \varepsilon$

Q4. GNF is useful because every derivation step produces exactly:

A) one non-terminal

B) two non-terminals

C) one terminal symbol

D) one terminal + one non-terminal

Thank You